

VLSI INTERNSHIP – TASK 3

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Domain: Very Large Scale Integration (VLSI) Intern

Goal:

Run basic STA (clock, setup/hold checks) on synthesized netlist using timing libraries or representative delays.

Requirements:

- Prepare timing constraints (SDC) — clocks, I/O delays
- Use PrimeTime / open-source alternatives for STA
- Identify timing violations and propose fixes (pipelining, re-synthesis)

After running the design in OpenLane, I obtained the following STA report.

```
=====
report_tns
=====
tns 0.00
=====

report_wns
=====
wns 0.00
=====

report_worst_slack -max (Setup)
=====
worst slack 5.07

report_worst_slack -min (Hold)
=====
worst slack 1.51
```

Interpretation:

Total Negative Slack(TNS) = 0.00

No timing violations in the design.

Worst Negative Slack(WNS) = 0.00

No negative slack paths.

Timing constraints are satisfied.

Worst Setup Slack = 5.07 ns

The slowest path still has 5.07 ns of margin before violating the clock period.

Setup timing is met comfortably.

Worst Hold Slack = 1.51 ns

Hold timing is also satisfied.

No hold violations.